

DonorHyperglycemiaStatsProject

Running Code

Hypothesis testing for relationship between donor demographics and delayed liver graft function. All tests are compared to a significance level of 0.05.

Donor Age Hypothesis Test

Welch Two Sample t-test

```
data: d_age by delayed_fn
t = -1.2831, df = 234.05, p-value = 0.2007
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -5.648353  1.192857
sample estimates:
mean in group 0 mean in group 1
      42.10329      44.33103
```

The null hypothesis is that the mean ages of both groups are equal, while the alternative hypothesis is that the mean ages are not equal. The test statistic follows a t distribution with 234.05 degrees of freedom. The p-value is 0.2007, which is above the significance level of 0.05. Therefore, we fail to reject the null hypothesis of the mean donor ages being the same, regardless of whether or not there was delayed liver graft function.

Sex Hypothesis Test:

The null hypothesis is that there is no relationship between sex and Delayed Liver Graft Function. The alternative hypothesis is that there is a relationship.

Pearson's Chi-squared test with Yates' continuity correction

```
data: table(DonorHyperglycemia$d_genderf, DonorHyperglycemia$delayed_fn)
X-squared = 0.47116, df = 1, p-value = 0.4925
```

The test statistic 0.47116 follows a Chi-squared distribution with one degree of freedom. The p-value of 0.5172 is greater than the significance level of 0.4925, so we fail to reject the null hypothesis that there is no relationship. There is insufficient evidence to support there being a relationship between donor gender and delayed liver graft function.

Cause of Death Hypothesis Test

The null hypothesis is that there is no relationship between donor cause of death and delayed liver graft function. The alternative hypothesis is that there is a relationship.

	0	1
1	179	72
2	152	43
3	84	22
4	11	8

Fisher's Exact Test for Count Data

```
data: table(DonorHyperglycemia$d_cod, DonorHyperglycemia$delayed_fn)
p-value = 0.08737
alternative hypothesis: two.sided
```

The p-value under a fisher's exact test is 0.08737 which is greater than the significance level of 0.05, therefore we fail to reject the null. There is insufficient evidence to support a relationship between donor cause of death and delayed liver graft function

Race Hypothesis Test

The null hypothesis is that there is no relationship between donor whiteness and delayed liver graft function. The alternative hypothesis is that there is a relationship.

Pearson's Chi-squared test with Yates' continuity correction

```
data: table(DonorHyperglycemia$d_caucasian, DonorHyperglycemia$delayed_fn)
X-squared = 0.0067798, df = 1, p-value = 0.9344
```

The test statistic follows a chi-squared distribution with one degree of freedom. The p-value of 0.9344 is greater than the significance level of 0.5, therefore we fail to reject the null hypothesis. There is insufficient evidence to support a relationship between donor whiteness and delayed liver graft function.

Recipient MELD Scores (Assigned and Calculated)

```
t.test(r_meld_calc ~ delayed_fn, data = DonorHyperglycemia)
```

Welch Two Sample t-test

```
data: r_meld_calc by delayed_fn
t = -1.3365, df = 233.1, p-value = 0.1827
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -2.8198807  0.5403924
sample estimates:
mean in group 0 mean in group 1
    18.00711      19.14685
```

```
t.test(r_meld_assign ~ delayed_fn, data = DonorHyperglycemia)
```

Welch Two Sample t-test

```
data: r_meld_assign by delayed_fn
t = 0.3125, df = 241.92, p-value = 0.7549
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -0.8728208  1.2019739
sample estimates:
mean in group 0 mean in group 1
    22.33976      22.17518
```

High Risk vs. Low Risk

```
chisq.test(table(DonorHyperglycemia$high_risk, DonorHyperglycemia$delayed_fn))
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: table(DonorHyperglycemia$high_risk, DonorHyperglycemia$delayed_fn)
X-squared = 4.6205, df = 1, p-value = 0.03159
```

HEMO INSTABILITY

```
table(DonorHyperglycemia$d_genderf, DonorHyperglycemia$hemo_instability)
```

	0	1
0	59	280
1	37	175

```
test_result <- chisq.test(table(DonorHyperglycemia$hemo_instability, DonorHyperglycemia$d_genderf))
print(test_result)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: table(DonorHyperglycemia$hemo_instability, DonorHyperglycemia$d_genderf)
X-squared = 5.1073e-30, df = 1, p-value = 1
```

```
table(DonorHyperglycemia$high_risk, DonorHyperglycemia$hemo_instability)
```

	0	1
no	57	292
yes	12	62

```
test_result <- chisq.test(table(DonorHyperglycemia$hemo_instability, DonorHyperglycemia$high_risk))
print(test_result)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: table(DonorHyperglycemia$hemo_instability, DonorHyperglycemia$high_risk)
X-squared = 3.0272e-30, df = 1, p-value = 1
```

Logistic Regression

```
m1 <- glm(delayed_fn ~ d_age + d_cod + d_caucasian + d_genderf + donorrisk,
          data = DonorHyperglycemia,
          family = "binomial")

summary(m1)
```

Call:

```
glm(formula = delayed_fn ~ d_age + d_cod + d_caucasian + d_genderf +
     donorrisk, family = "binomial", data = DonorHyperglycemia)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.106469	0.628056	-3.354	0.000797	***
d_age	-0.005237	0.008304	-0.631	0.528224	
d_cod2	-0.312951	0.301977	-1.036	0.300045	
d_cod3	-0.255460	0.324765	-0.787	0.431517	
d_cod4	0.763904	0.565541	1.351	0.176776	
d_caucasian	-0.013151	0.274557	-0.048	0.961797	
d_genderf	-0.094485	0.247659	-0.382	0.702822	
donorrisk	1.046880	0.403866	2.592	0.009538	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 497.37 on 429 degrees of freedom
Residual deviance: 482.50 on 422 degrees of freedom
```

(142 observations deleted due to missingness)
AIC: 498.5

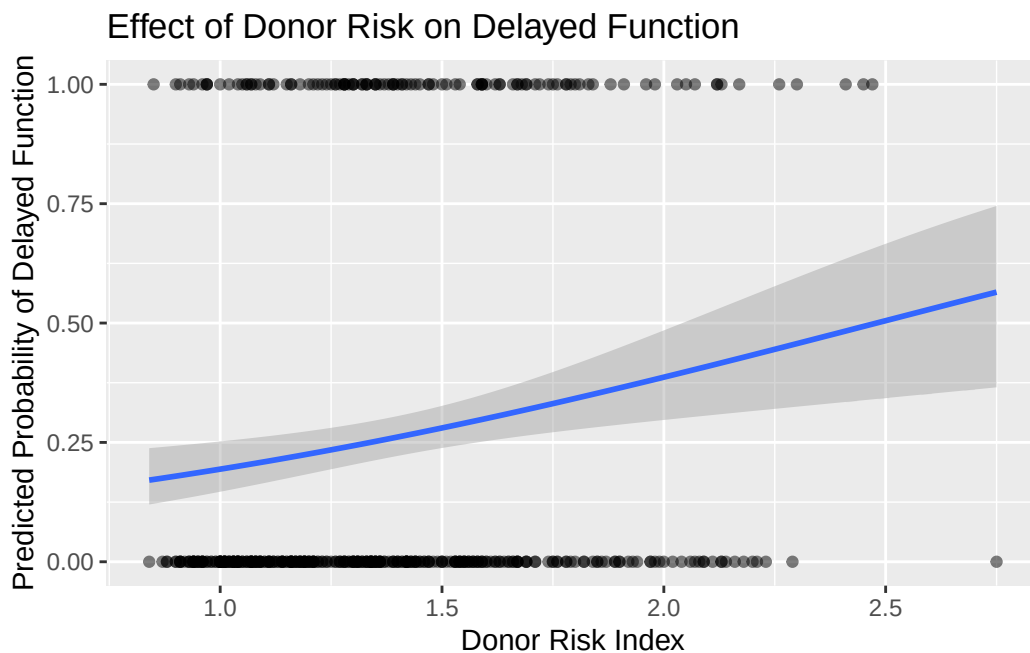
Number of Fisher Scoring iterations: 4

```
library(ggplot2)
ggplot(DonorHyperglycemia, aes(x = donorrisk, y = as.numeric(delayed_fn))) +
  geom_point(alpha = 0.5) +
  stat_smooth(method = "glm",
              method.args = list(family = "binomial"),
              se = TRUE) +
  labs(title = "Effect of Donor Risk on Delayed Function",
       x = "Donor Risk Index",
       y = "Predicted Probability of Delayed Function")
```

`geom\_smooth()` using formula = 'y ~ x'

Warning: Removed 135 rows containing non-finite outside the scale range
(`stat\_smooth()`).

Warning: Removed 135 rows containing missing values or values outside the scale range
(`geom\_point()`).

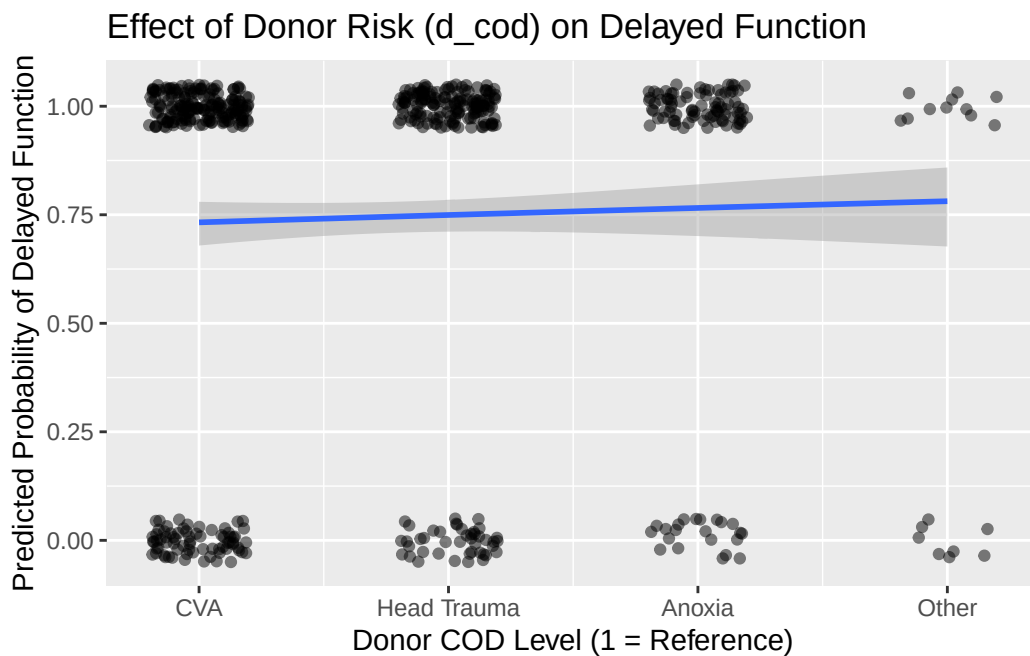


```
library(ggplot2)
ggplot(DonorHyperglycemia, aes(x = as.numeric(d_cod), y = as.numeric(delayed_fn != 1))) +
  geom_jitter(height = 0.05, width = 0.2, alpha = 0.5) +
  stat_smooth(method = "glm",
              method.args = list(family = "binomial"),
              se = TRUE) +
  scale_x_continuous(breaks = 1:4, labels = c("CVA" , "Head Trauma", "Anoxia" , "Other")) +
  labs(title = "Effect of Donor Risk (d_cod) on Delayed Function",
       x = "Donor COD Level (1 = Reference)",
       y = "Predicted Probability of Delayed Function")
```

`geom\_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat\_smooth()`).

Warning: Removed 1 row containing missing values or values outside the scale range
(`geom\_point()`).



```
library(ggplot2)
ggplot(DonorHyperglycemia, aes(x = d_cod, y = as.numeric(delayed_fn) - 1)) +
  geom_jitter(alpha = 0.5, height = 0.05, width = 0.2) +
  stat_smooth(aes(group = 1),
              method = "glm",
              method.args = list(family = "binomial"),
              se = TRUE) +

  labs(title = "Effect of Donor Risk on Delayed Function",
       x = "Cause of Death",
       y = "Predicted Probability of Delayed Function")
```

`geom\_smooth()` using formula = 'y ~ x'

Warning: Failed to fit group 1.

Caused by error:

! y values must be 0 <= y <= 1

